ABHINAV PANDEY

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PROFESSIONAL SUMMARY

B.Tech (Computer Science) graduate with hands-on experience across the full software development life cycle (SDLC) - functional and technical design, coding, unit testing, and system integration - for full-stack web applications built with Java, Spring Boot, JavaScript, HTML, and CSS. Designs and develops components for custom information systems using microservices architecture, relational database design (MySQL), and secure authentication/authorization (JWT, Spring Security) following OWASP best practices. Applies data structures and algorithmic thinking to debug service-layer issues and resolve performance and concurrency problems, and documents designs, code, and test cases to support long-term maintainability. Uses AI-assisted development tools (GitHub Copilot, ChatGPT) to speed up routine coding and testing tasks. Proficient in API testing with Postman and unit/integration testing with JUnit and Mockito, applying established coding best practices and testing frameworks. Built ChatNova, a real-time communication platform, as a B.Tech capstone project. Self-motivated and collaborative, with working knowledge of Agile methodology and eagerness to learn from senior engineers while contributing to high-quality, scalable information systems.

TECHNICAL SKILLS

Languages: Java (Java 8+, Streams, Lambdas, Multithreading), SQL, JavaScript (Frontend & Backend Integration), HTML5, CSS3

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, JDBC, Lombok

API & Web: RESTful API Design, WebSocket (STOMP), API Gateway, JSP

Auth & Security: JWT (JSON Web Tokens), Spring Security, Role-Based Access Control (RBAC), BCrypt, OWASP Awareness

Databases: MySQL (Relational Schema Design & Query Optimization), Redis (Caching & Session Management)

Architecture: Microservices Architecture, Eureka Service Discovery, API Gateway, Event-Driven Design, Design Patterns (Singleton, Factory, Builder)

DevOps & Tools: Docker, Git, GitHub, Maven, Postman, Eclipse, Spring Tool Suite (STS), Apache Tomcat

Testing: JUnit, Mockito (Unit & Integration Testing), API Testing with Postman

Practices & Methodologies: SDLC, Agile (Scrum), Waterfall (academic exposure), Component & Technical Design, Technical Documentation, System Integration, AI-Assisted Development (GitHub Copilot, ChatGPT)

CS Fundamentals: Data Structures & Algorithms (DSA), OOPs Principles, DBMS

PROJECT EXPERIENCE

ChatNova - Real-Time Chat & Messaging Platform | *B.Tech Final Year Major Project* | 2025 - 2026

Java, Spring Boot, Spring Security, JWT, WebSocket, Redis, MySQL, Microservices, JUnit, Mockito, HTML5, CSS3, JavaScript

ChatNova is a real-time messaging platform built for my final year project, aimed at understanding how apps like WhatsApp or Slack handle live messaging, authentication, and scale behind the scenes. It lets users sign up securely and chat one-on-one or in groups with live delivery/seen status, all backed by a microservices architecture I designed from scratch.

- Produced the technical design for a microservices-based system and assembled its components - API Gateway and Eureka Service Discovery - into a cohesive architecture supporting dynamic service registration, centralized routing, and horizontal scalability.
- Implemented JWT-based authentication and Spring Security (RBAC) following OWASP best practices; built real-time private/group messaging via WebSocket (STOMP) with full message-status tracking (Sent, Delivered, Seen) using Java 8 Streams and Lambdas.
- Designed and documented a relational MySQL schema via Spring Data JPA, integrated Redis for caching, and built group-chat creation and participant-management modules with a responsive HTML/CSS/JavaScript frontend

(Builder/Factory patterns); validated functionality with JUnit + Mockito tests, centralized exception handling, and technical documentation for future maintenance.

Smart Task Management System | *Personal Project* | 2024

Java, Spring Boot, Spring Security, JWT, MySQL, Multithreading, JUnit, Mockito, HTML, CSS, JavaScript

Smart Task Management System is a personal productivity app that lets users create, prioritize, and track tasks with deadlines. I built it to get hands-on experience designing a complete backend on my own, from secure authentication to database design to handling background notifications, instead of just following a tutorial.

- Produced the functional and technical design for a scalable RESTful task-management API in Spring Boot - covering task creation, prioritization, deadline tracking, and soft-delete - applying data structures and algorithmic problem-solving for efficient prioritization logic and the Singleton pattern for service-layer components.
- Implemented JWT authentication and RBAC via Spring Security with BCrypt hashing; leveraged Java Multithreading for async notification dispatch, Spring Data JPA/MySQL persistence, a responsive HTML/CSS/JavaScript dashboard, and JUnit + Mockito tests covering critical service methods.

Enterprise Banking System | *Java Backend Developer Project* | 2026

Java 21, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, Spring Cloud (API Gateway, Eureka), Redis, MySQL, JUnit, Mockito

Enterprise Banking System is a backend I built to simulate how a real banking platform works, covering customer onboarding, account management, beneficiary handling, and fund transfers, with the security and reliability that real financial systems demand.

- Designed and assembled a secure, scalable microservices-based Enterprise Banking System simulating real-world banking operations, with RESTful APIs for customer onboarding, account management, beneficiary management, fund transfers, and transaction history.
- Implemented JWT-based authentication and Spring Security (RBAC) for Admin and Customer roles, securing all financial transaction endpoints; integrated Redis for caching and used Spring Cloud API Gateway/Eureka for scalable inter-service communication.
- Persisted entities in MySQL via Spring Data JPA/Hibernate with centralized exception handling and input validation; wrote JUnit + Mockito unit tests and maintained technical documentation of components and test cases to ensure code quality, support future maintenance, and reduce regression risk.

EDUCATION

Bachelor of Technology (B.Tech) - Computer Science & Engineering (Completed) / *2022 - 2026*
Gurukula Kangri (Deemed to be University), Haridwar

Higher Secondary (Class XII) - Science / *2022*
Uttar Pradesh Board | Percentage: 81.4%

High School (Class X) / *2020*
Uttar Pradesh Board | Percentage: 87.5%

ACHIEVEMENTS & ACTIVITIES

- Completed self-directed learning in Docker, Java 8+ Streams/Lambdas, and Mockito testing, applying Agile-style iterative practices; regularly practices DSA problem-solving to strengthen analytical thinking and write efficient, well-documented code.

SOFT SKILLS

Problem-Solving, Troubleshooting, Communication, Analytical Thinking, Adaptability